

Embedded Systems and STM32F4 (Cortex-M4) perspective 📌

🧠 1. Processor Core vs Processor

Aspect	Processor Core	Processor (CPU)
Definition	The <i>central computation engine</i> that executes instructions (e.g., Cortex-M4).	The <i>complete processing unit</i> that includes one or more cores plus peripherals.
Components	ALU, Registers, Control Unit, Pipeline, NVIC, SysTick, etc.	Processor core + Bus interfaces + Memory system + Debug units + Timers, ADC, communication interfaces.
Example	ARM Cortex-M4 core.	STM32F407 microcontroller processor (includes Cortex-M4 core + peripherals).
Role	Executes the instruction set (e.g., ARM Thumb-2 ISA).	Provides the platform for application execution with memory and I/O.

➡ **In STM32F4:** The *processor core* is ARM Cortex-M4; the *processor* is the overall CPU inside the STM32 MCU that integrates this core with other modules.

⚙️ 2. Processor vs Microcontroller

Aspect	Processor (CPU)	Microcontroller (MCU)
Definition	Only the core that executes code. Needs external memory and I/O.	A complete embedded system on a single chip.
Includes	ALU, Control Unit, Registers.	Processor + Flash + SRAM + Timers + ADC + GPIO + UART/SPI/I2C + Interrupt Controller.
Usage	PCs, laptops, high-performance systems.	Embedded systems, control systems, IoT, robotics.
External Dependencies	Needs external RAM, ROM, peripherals.	Self-contained (integrated memory and peripherals).
Example	Intel i7, ARM Cortex-A72.	STM32F407, ATmega328P, ESP32.

➡ STM32F4 = Microcontroller

It contains the **Cortex-M4 processor core** inside, plus Flash, SRAM, timers, UARTs, GPIO, etc.

🌱 3. Features of Cortex-Mx Processor (focus: Cortex-M4)

Feature	Description
Architecture	32-bit ARMv7-M (Harvard architecture).
Instruction Set	Thumb-2 (mix of 16-bit and 32-bit instructions for high performance + compact code).
Pipeline	3-stage pipeline (Fetch–Decode–Execute).
NVIC	Nested Vectored Interrupt Controller — supports 240 interrupts with programmable priorities.
SysTick Timer	Dedicated timer for OS tick or time base generation.
FPU	Optional single-precision Floating Point Unit (in Cortex-M4F).
MPU	Optional Memory Protection Unit for memory region access control.
Low-Power Modes	Sleep, Deep Sleep, Stop, Standby.
Exception Model	Supports system exceptions (Reset, NMI, HardFault, etc.).
Bus Interface	AHB, APB (Advanced High-Performance/Peripheral Bus).
Debug Support	JTAG/SWD (Serial Wire Debug), breakpoint & watchpoint support.
DSP Instructions	Multiply-Accumulate (MAC), SIMD for signal processing.

🌀 4. Operational Modes of Cortex-Mx Processor

Cortex-M processors operate in **two modes** only:

Mode	Description	Entry Condition
Thread Mode	Normal program execution mode (main application code or OS task).	After reset, or after exiting from an exception.

Mode	Description	Entry Condition
Handler Mode	When executing an exception or interrupt service routine (ISR) .	When an interrupt or fault occurs.

➔ **Example (STM32F4):**

- Main() executes in **Thread mode**.
- When SysTick interrupt occurs, ISR runs in **Handler mode**.

5. Access Levels of the Processor

Cortex-M supports **two privilege levels**:

Access Level	Access Rights	Used For
Privileged	Full access to all registers, memory regions, and special instructions.	OS kernel, device drivers, interrupt handlers.
Unprivileged	Restricted access — can't modify control registers or protected memory.	Application or user-level tasks.

➔ **Switching Access Levels:**

- Controlled by the **CONTROL** register.
- OS (like FreeRTOS) often uses **privileged mode** for kernel and **unprivileged mode** for user tasks for safety.

6. Memory-Mapped and Non-Memory-Mapped Registers

a. Memory-Mapped Registers

- In ARM Cortex-M systems, **most peripherals and core registers are memory-mapped**.
- Each peripheral (GPIO, USART, ADC, etc.) is assigned a **fixed memory address range**.
- The CPU accesses them using **load/store (LDR/STR)** instructions just like normal memory.

Example (STM32F4)	Address	Description
GPIOA_MODER	0x40020000	GPIOA mode configuration register.

Example (STM32F4)	Address	Description
USART1_DR	0x40011004	USART1 data register.
RCC_AHB1ENR	0x40023830	Clock enable register for AHB1 peripherals.

➔ Usage:

```
#define RCC_AHB1ENR (*(volatile uint32_t*)0x40023830)

RCC_AHB1ENR |= (1 << 0); // Enable GPIOA clock
```

b. Non-Memory-Mapped Registers

- Accessed via **special instructions**, not through memory addresses.
- Examples include **core system registers** like:
 - **CONTROL** – selects Thread/Handler mode and privilege level.
 - **PRIMASK, FAULTMASK, BASEPRI** – interrupt masking.
 - **MSP, PSP** – stack pointers.
 - **APSR, IPSR, xPSR** – program status registers.

➔ These are **special registers** accessed using MRS/MSR assembly instructions (not normal memory).

7. Summary (STM32F4 – Cortex-M4 Context)

Category	Key Points
Core	ARM Cortex-M4F (32-bit, Thumb-2, DSP + FPU).
Modes	Thread mode, Handler mode.
Access Levels	Privileged / Unprivileged.
Memory System	Harvard architecture with AHB/APB buses.
Interrupts	NVIC with priority levels and vector table.
Registers	Mix of memory-mapped (peripheral) and core (non-memory-mapped).
Debugging	SWD/JTAG, breakpoint, and trace support.